

Lexical Analysis Automata Execution Trace Report

Generated inside compiler1 workspace via integrated pipeline

Target Input Program Stream Evaluated:

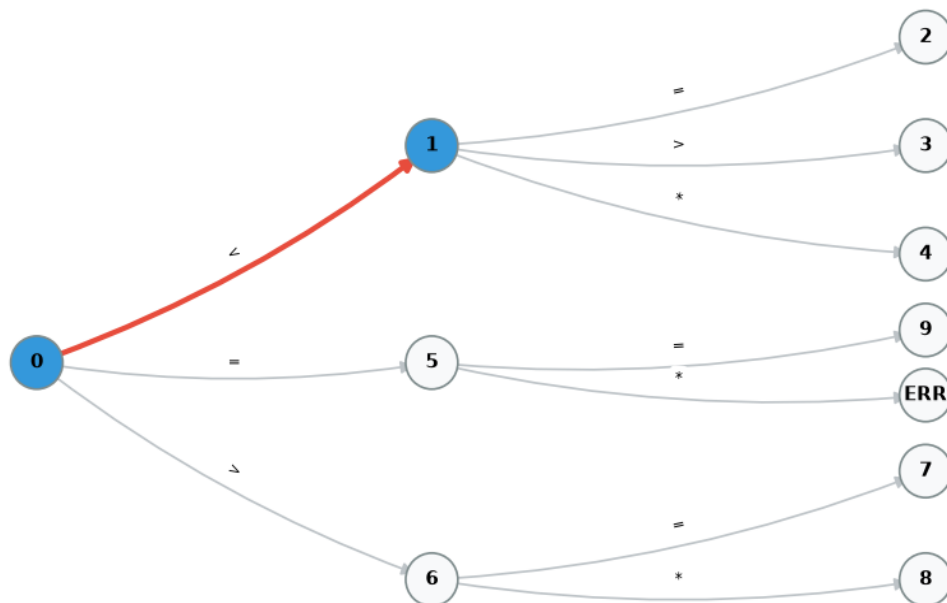
<= <> < == >= > = <=) (

Step 1: Lexeme Transaction Trace

Log Description: < (State 0 -> 1 -> Retract): Reads < then looks ahead at the space ' '. Because space doesn't match = or >, it retracts the index by 1 so the space isn't lost, drops to State 4, and returns LT.

DFA Traveled State Sequence Path: 0 → 1 → Retract

■■ Input Pushback (Retraction) Triggered: Lexer engine encountered a structural code delimiter lookahead constraint. Stream pointer retracted by 1 step safely.

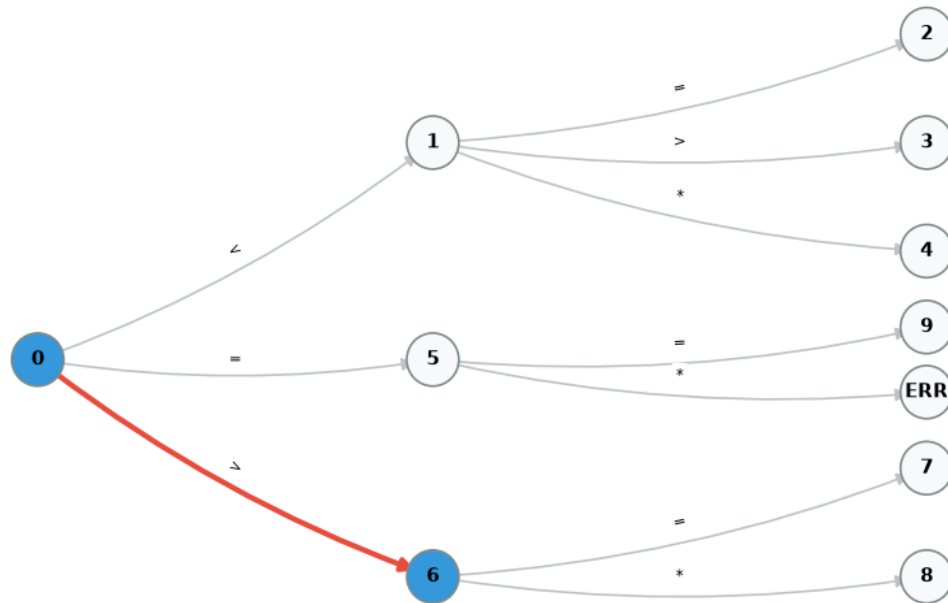


Step 2: Lexeme Transaction Trace

Log Description: > (State 0 -> 6 -> Retract): Reads > and looks ahead at the space '>'. It retracts the index back by 1, drops to State 8, and returns GT.

DFA Traveled State Sequence Path: 0 → 6 → Retract

■■ Input Pushback (Retraction) Triggered: Lexer engine encountered a structural code delimiter lookahead constraint. Stream pointer retracted by 1 step safely.

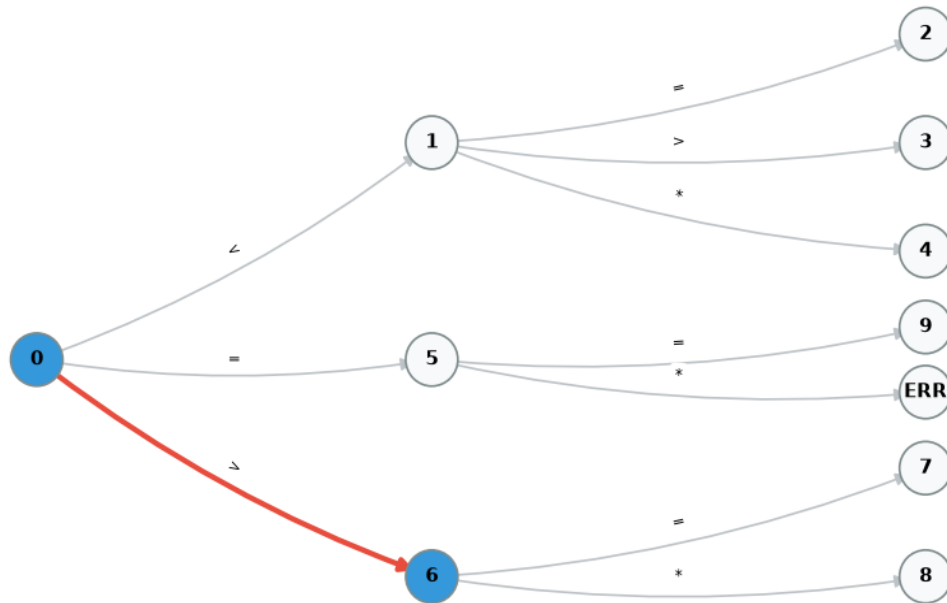


Step 3: Lexeme Transaction Trace

Log Description: > (State 0 -> 6 -> Retract): Reads > and looks ahead at the space ' '. It retracts the index back by 1, drops to State 8, and returns GT.

DFA Traveled State Sequence Path: 0 → 6 → Retract

■■ Input Pushback (Retraction) Triggered: Lexer engine encountered a structural code delimiter lookahead constraint. Stream pointer retracted by 1 step safely.



Step 4: Lexeme Transaction Trace

Log Description: <= (State 0 -> 1 -> 2): Consumes two characters, matches perfectly, returns LE.

DFA Traveled State Sequence Path: 0 → 1 → 2

Pushback Status: None (Direct terminal state validation match)

